Even and Odd Functions

1 Background Knowledge Check

1. **Evaluating functions**
   Any function such as $f(x) = x^2$ connects an independent variable (in this case, x) with a dependent variable ($f(x)$). The function can be used to find the dependent variable ($f(x)$) at any x-value. If you want to know what the value of the function is when $x = 2$ in $f(x) = x^2$, we simply plug in 2 for x:

$$f(2) = 2^2$$

$$f(2) = 4$$

Therefore, when $x = 2$, $f(2) = 4$:

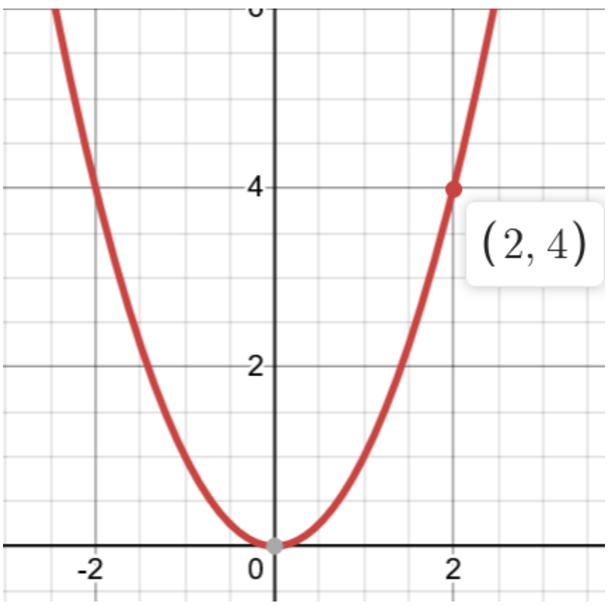

Figure 1: $f(x) = x^2$

2. **Graphing functions**
   When working with even and odd functions, it is important to be able to graph, or at the very least, visualize how a function's graph may look from the equation itself. This requires being familiar with graphs of base functions such as $y = x$, $y = x^2$, $y = e^x$, $y = \sin(x)$, $y = \frac{1}{x}$, etc.

3. **Working with Negative Values**
   Many functions, such as $f(x) = x^2$, can be evaluated at negative values of x.

   For example, at $x = -2$:

$$f(-2) = (-2)^2$$

$$f(-2) = 4$$

Notice that this is the same value we obtain when $x = 2$.

However, this is not always the case. Consider the function $f(x) = x^3$.

At $x = 2$:

$$f(2) = 2^3 = 8$$

At $x = -2$:

$$f(-2) = (-2)^3 = -8$$

In this case, the output changes sign.

Understanding how functions behave at negative values of x is important when classifying functions as even or odd.

2 Even Functions

Even functions are functions that are symmetric about the y-axis. To graphically check if any function is even, imagine taking the graph on the positive side of the y-axis and mirroring it to the negative side. If the mirrored image matches the actual graph, the function is even.
Here is an example with the graph of $f(x) = \cos(x)$: Starting with the main graph:

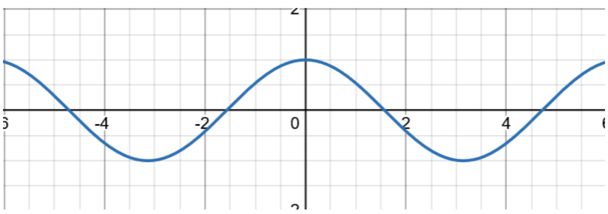

Figure 2: $f(x) = \cos(x)$

Let's focus on the positive side:

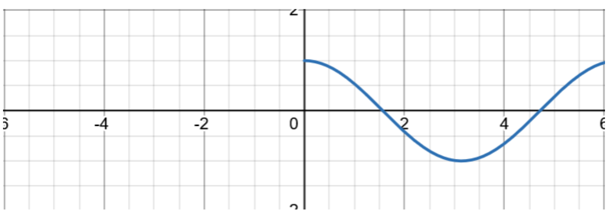

Figure 3: $f(x) = \cos(x)\{x \geq 0\}$

Now, let's imagine that the y-axis is a mirror. Mirroring this to the negative side of the y-axis gives:

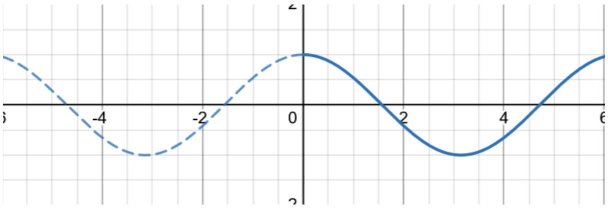

Figure 4: $f(x) = \cos(x)\{x \geq 0\}$ reflected across the y-axis

Since this mirrored graph matches our original graph for $f(x) = \cos(x)$, we can conclude that $f(x) = \cos(x)$ is an even function.

3 Odd Functions

Similar to even functions, odd functions also showcase symmetry. An odd function is one that is symmetric about the origin. To check if a function is odd, we take the part of the graph on the positive side of the y-axis and rotate it 180° about the origin.
Let's look at the graph of $\sin(x)$:

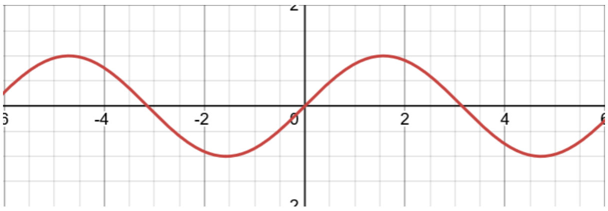

Figure 5: $f(x) = \sin(x)$

Now, lets consider just the part of the graph on the right side of the y-axis:

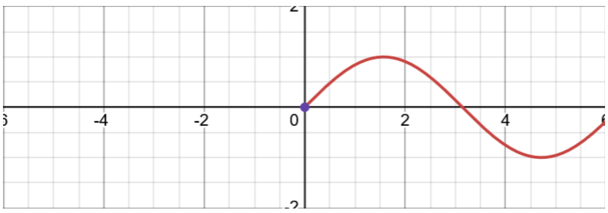

Figure 6: $f(x) = \sin(x)\{x \geq 0\}$

Let's imagine taking this graph and rotating it 180° about the origin (purple dot):

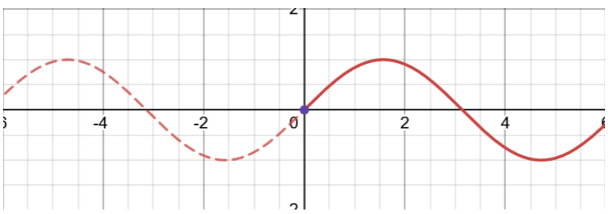

Figure 7: $\sin(x)\{x \geq 0\}$ rotated 180° about the origin

Since this graph perfectly matches our original graph for $\sin(x)$, we can conclude that $f(x) = \sin(x)$ is an odd function. In the graph above, every point that makes up the

graph on the right side of the y-axis has its x and y value multiplied by -1 in order to form the dotted graph on the left side of the y-axis.
The following image shows the original point $(\frac{\pi}{2}, 1)$ (green) and the position where this point is rotated to $(-\frac{\pi}{2}, -1)$ (blue):

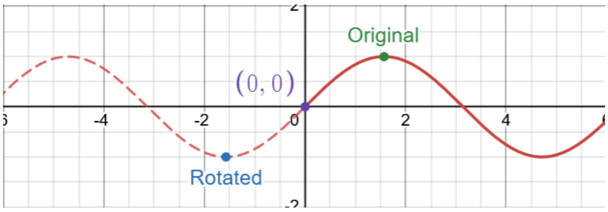

Figure 8: The original and rotated point

Since the original point was rotated 180° about the origin, a line going through both the rotated and original point would also go through the origin.

Each point (x, y) is mapped to $(-x, -y)$ during a 180° rotation about the origin.

4 Classifying Functions Mathematically

Often times, we may be asked to or want to classify functions as odd or even mathematically instead of graphically.

1. **Even Functions:**
   A function is even if $f(-x) = f(x)$. This makes sense as from what we saw before, a function is even if it has the same value for both positive and negative inputs, resulting in symmetry about the y-axis. Let's apply this to $f(x) = x^2$:
   $$f(x) = x^2$$
   $$f(-x) = (-x)^2 = x^2 = f(x)$$
   $$f(-x) = f(x)$$
   Therefore, $f(x) = x^2$ is an even function.

2. **Odd Functions:**
   A function is odd if $f(-x) = -f(x)$. This makes sense as for an odd function, every (x, y) value pair has a corresponding $(-x, -y)$ pair, resulting in symmetry about the origin. Let's apply this to $f(x) = x^3$:
   $$f(x) = x^3$$
   $$f(-x) = (-x)^3 = -x^3 = -f(x)$$
   $$f(-x) = -f(x)$$
   Therefore, $f(x) = x^3$ is an odd function.

Therefore, to classify a function as odd, even or neither, we evaluate $f(-x)$ and compare it to $f(x)$.

5 Conclusion

Even Functions:

- An even function is a function that is symmetrical about the y-axis.
- Graphically, this can be checked by taking the part of the function on the right side of the y-axis, reflecting it over the y-axis and checking if the reflection matches the part of the graph on the left side of the y-axis.
- Being symmetrical about the y-axis also means that the function outputs the same value for both positive and negative input values. Therefore, a function is even if it satisfies $f(-x) = f(x)$.

Odd Functions:

- Similarly, an odd function is a function that is symmetrical about the origin.
- Graphically, this can be checked by taking the part of the function on the right side of the y-axis and rotating it 180° around the origin and checking if the rotated graph matches the part of the graph on the left side of the y-axis.
- Being symmetrical about the origin also means that for any point (x, y), there also exists a point $(-x, -y)$. Therefore, a function is odd if it satisfies $f(-x) = -f(x)$

Therefore, mathematically, any function can be tested for being even, odd or neither by comparing $f(-x)$ to $f(x)$.

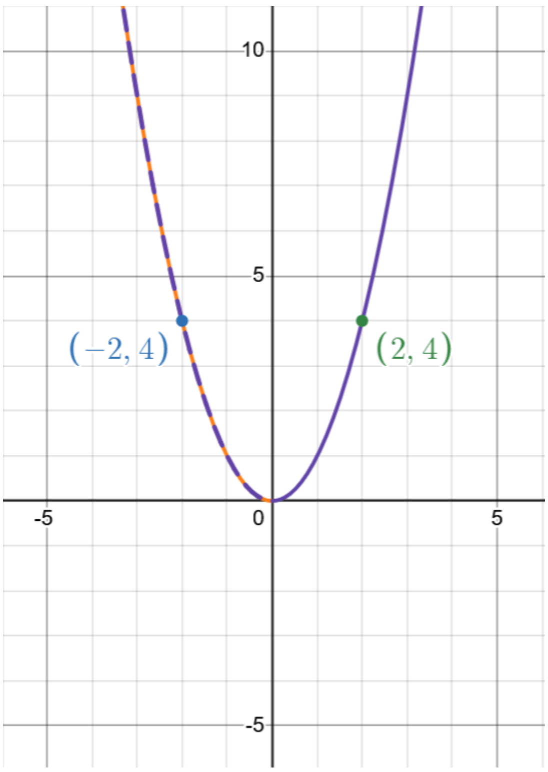

Figure 9: Even function: $f(x) = x^2$

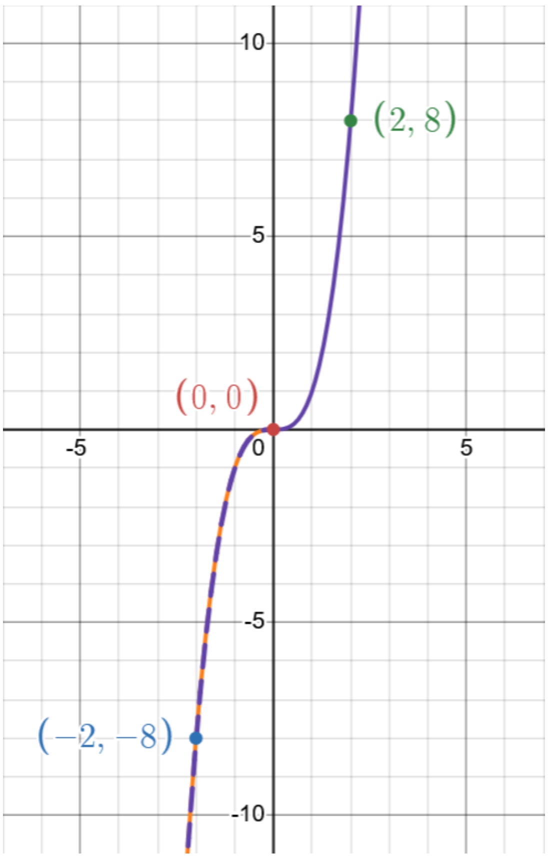

Figure 10: Odd function: $f(x) = x^3$